

KAIST AI COMPUTING

AI Computing

Nurturing AI-Native leaders who create AI and design advanced systems.

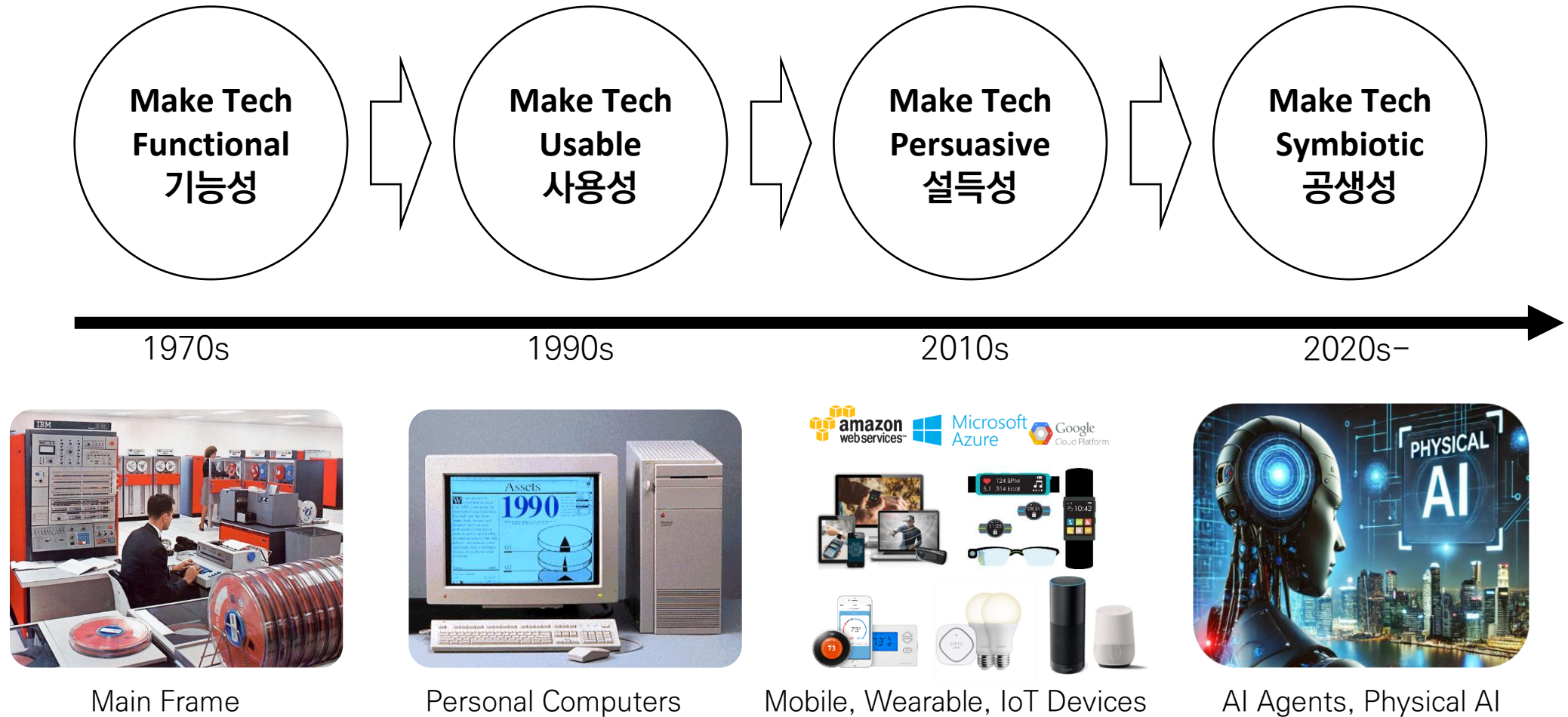
[Learn More](#) ↓[Admissions 2026](#)

Information Session
May 26, 2026

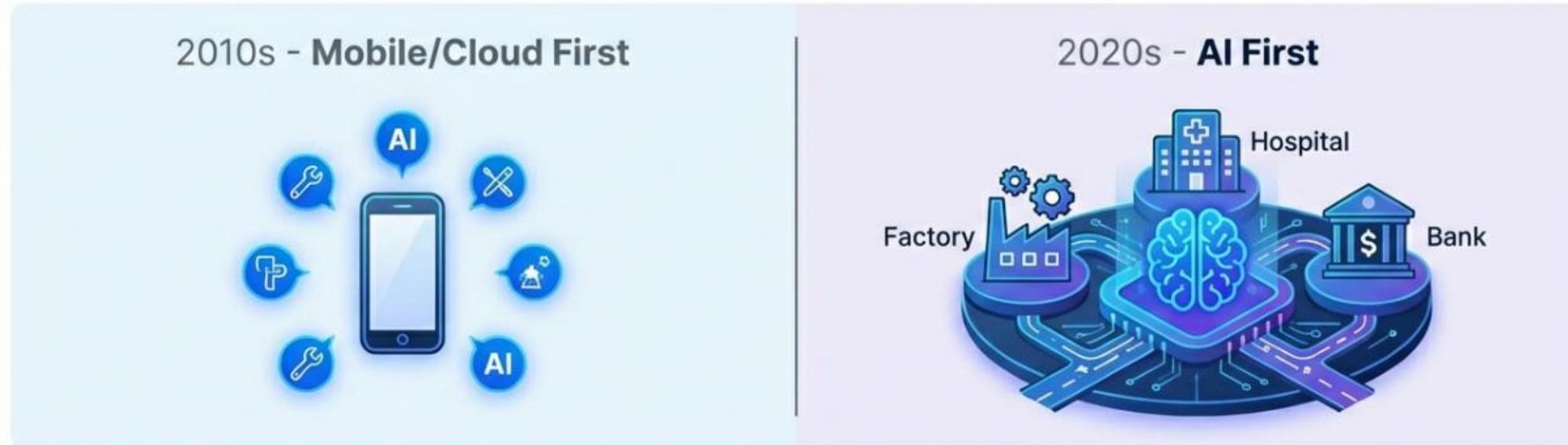
<https://aic.kaist.ac.kr>



Computing Trend

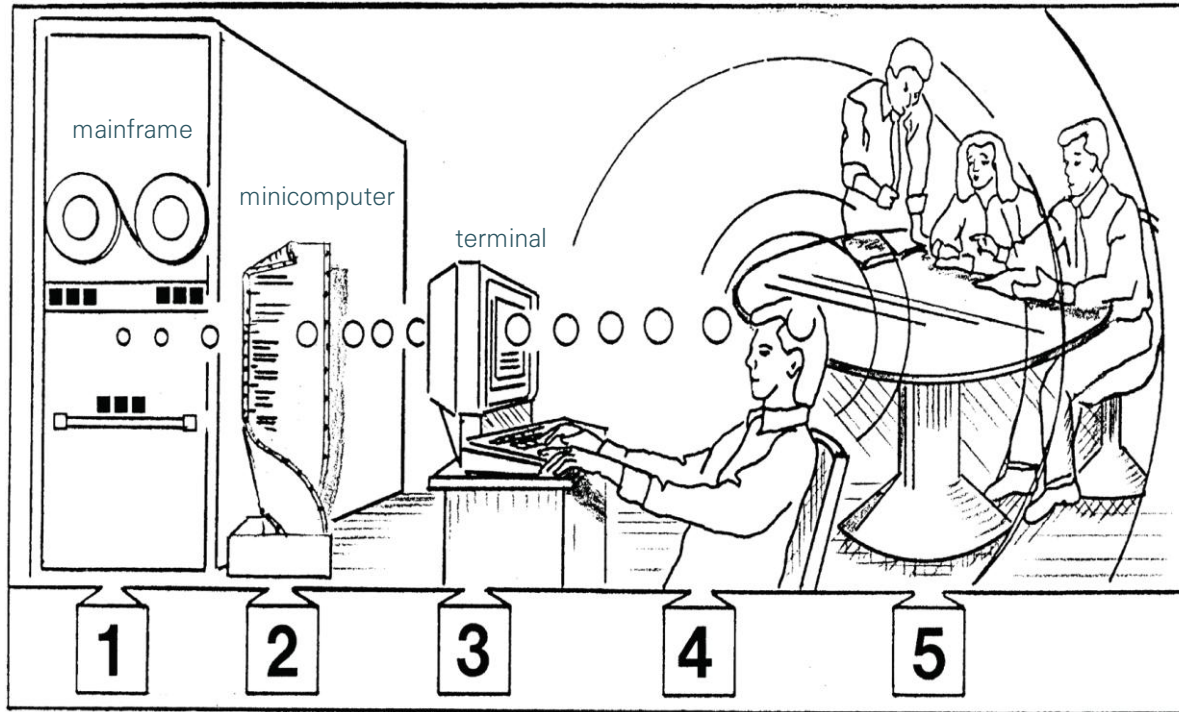


Industry Shifts: Mobile to AI



AI-Driven Industry	New Competitive Paradigm	National AI Assets
AI-centered redesign of industrial value chains	From AI adoption → AI internalization & design	Infrastructure · Algorithms · Data

Computing Paradigm Shift: From Tools to Partners



1
Interface as Hardware
인터페이스: 하드웨어

2
Software
소프트웨어

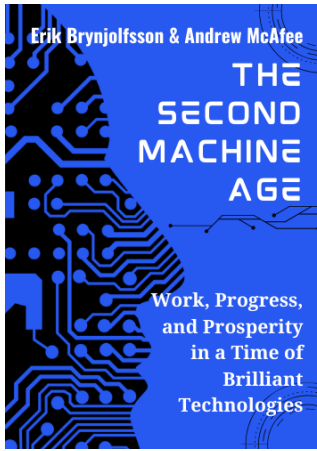
3
Terminal
터미널

4
Dialogue
대화

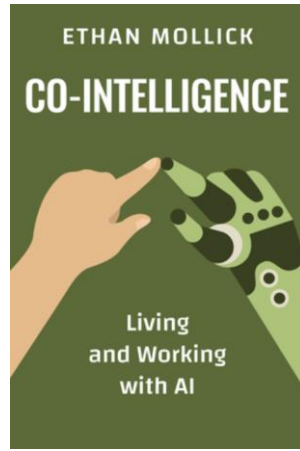
5
Social & Ubiquitous
소셜 및 유비쿼터스



From Tool to Partner: The Evolution of Human-Computer Interaction, Jonathan Grudin, 2018



Erik Brynjolfsson
Andrew McAfee

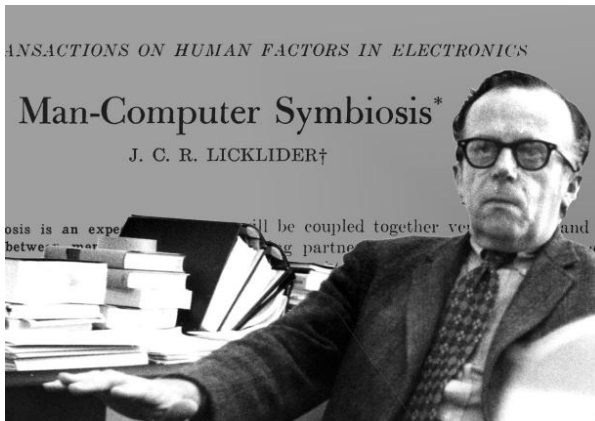


Ethan
Mollick

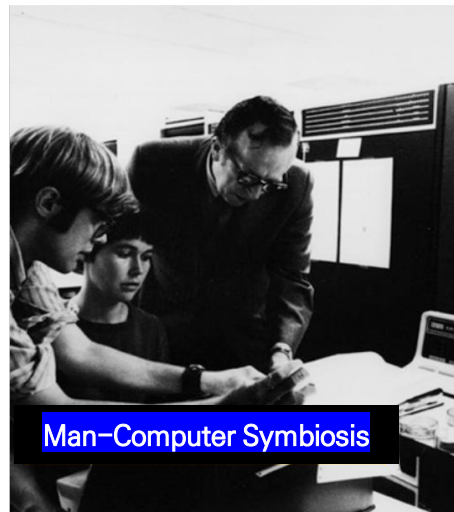
Automation in the Machine Age: In the machine age, many routine and repetitive tasks are automated (e.g., autonomous vehicles, question answering, personal assistants).

Beyond Routine Tasks: Humans are capable of creative idea generation, broad reasoning, and complex communication.

Co-Intelligence (Dual Brain): Through human-machine collaboration, limitations can be overcome; e.g., Human physician + AI agent



Man-Computer Symbiosis
J.C.R. Licklider 1960



“In the anticipated partnership between humans and computers, humans will set goals, formulate hypotheses, determine criteria, and perform evaluations. Computing machines will carry out the routine operations necessary for technical and scientific thinking. Preliminary analysis suggests that such a cooperative partnership will perform intellectual work much more effectively than humans alone.”

SW Ate the World. AI Will Eat SW.

Why Is Software Eating the World?

General

Why Software Is Eating the World

2011

Marc Andreessen
Posted August 20, 2011

Available in ▾

Share

Software is eating the world.

More than 10 years after the peak of the 1990s dot-com bubble, a dozen or so new Internet companies like Facebook and Twitter are sparking controversy in Silicon Valley, due to their rapidly growing private market valuations, and even the occasional successful IPO. With scars from the heyday of Webvan and Pets.com still fresh in the investor psyche, many investors, along with others, have been arguing the other side of the case. (I am co-founder and general partner of venture capital firm Andreessen Horowitz, which has invested in several of these companies, and I am also a personal investor in LinkedIn.) We believe that many of the prominent new Internet companies are building real, high-growth, high-margin, highly defensible businesses. Today's stock market actually rates technology, as shown by all-time low price/earnings ratios for major public technology companies. Apple, for example, has a P/E ratio of around 15.2—about the same as the broader stock market, despite Apple's immense success. Microsoft and Google have also become the biggest companies in America, judged by market capitalization, surpassing Exxon Mobil). And, perhaps most telling, you can't have a bubble when people are constantly screaming "Bubble!"

- Software became the core competitive force across industries
- Platform companies rapidly transformed traditional businesses
- Data and network effects strengthened market dominance
- The era of "Software is Eating the World" emerged

Industries were reorganized around software

Good News: AI Will Eat Application Software

Good news: AI Will Eat Application Software

2026

Alex Immerman and Santiago Rodriguez
Posted March 2, 2026

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The moats that matter aren't going away

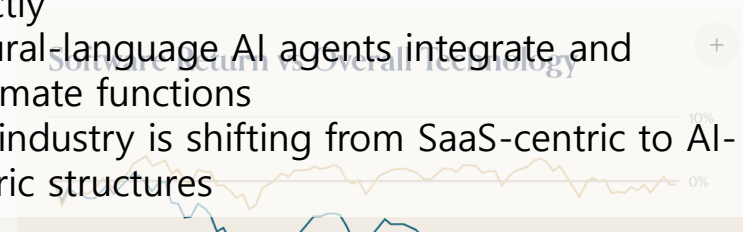
Platform shifts create new winners—and new moats

The great software bifurcation is coming

The software industry is having a panic attack. Since the start of 2026, ETFs for public software companies have fallen by 30 percent, erasing all the gains since the launch of ChatGPT. Companies like Salesforce, Adobe, Intuit, ServiceNow, and Veeva—bellwethers that have compounded investors' capital for a decade or more—are down 25 to 30 percent. They're calling it the "SaaSocalypse." It's rapidly become the market consensus: *AI is hollowing out the enterprise software customer base for enterprise software is hollowed out and the S&P enters a massive years-long drawdown.*

- AI is transforming the interface layer of software itself
- Users no longer navigate complex applications directly
- Natural language AI agents integrate and automate functions
- The industry is shifting from SaaS-centric to AI-centric structures

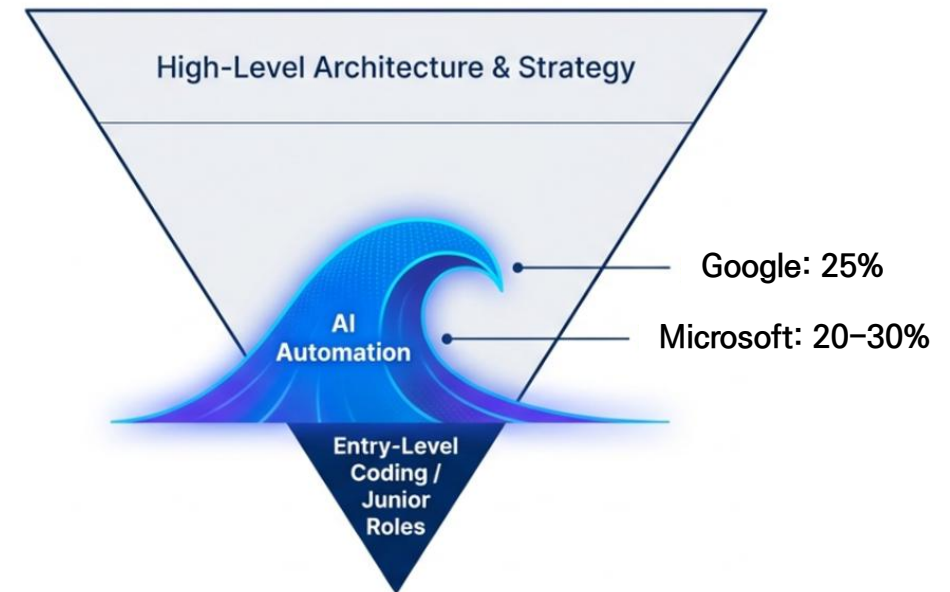
Software return vs. overall technology



AI is not merely replacing software, but expanding into a larger abstraction layer

AI is not ending programming — it is reinventing it

- Every major abstraction shift in history created more programmers, not fewer
- AI enables natural-language coding, but **expertise is still required** for design, debugging, and edge cases
- The **role shifts** from **writing code** to **systematic thinking and managing AI systems**
- The future belongs to developers who collaborate with AI, expanding what is programmable



The End of Programming as We Know It, By Tim O'Reilly February 4, 2025
<https://www.oreilly.com/radar/the-end-of-programming-as-we-know-it>

New Education Demand

Expanding AI Education

Growing industry demand and investment are driving universities to expand AI education

Rapid Growth of AI Majors

At leading universities, such as MIT, AI has become one of the fastest-growing majors

AI Undergraduate Programs in Korea

New AI undergraduate programs (e.g., KAIST, Yonsei, Korea University)

The New York Times

Grok Controversy Lawsuit Over A.I. Hiring Decisions Electricity Costs Data Centers in Space? Kille

College Students Flock to a New Major: A.I.

At M.I.T., a new program called “artificial intelligence and decision-making” is now the second-most-popular undergraduate major.

By [Natasha Singer](#)

Natasha Singer covers computer science and A.I. education.

Dec. 1, 2025

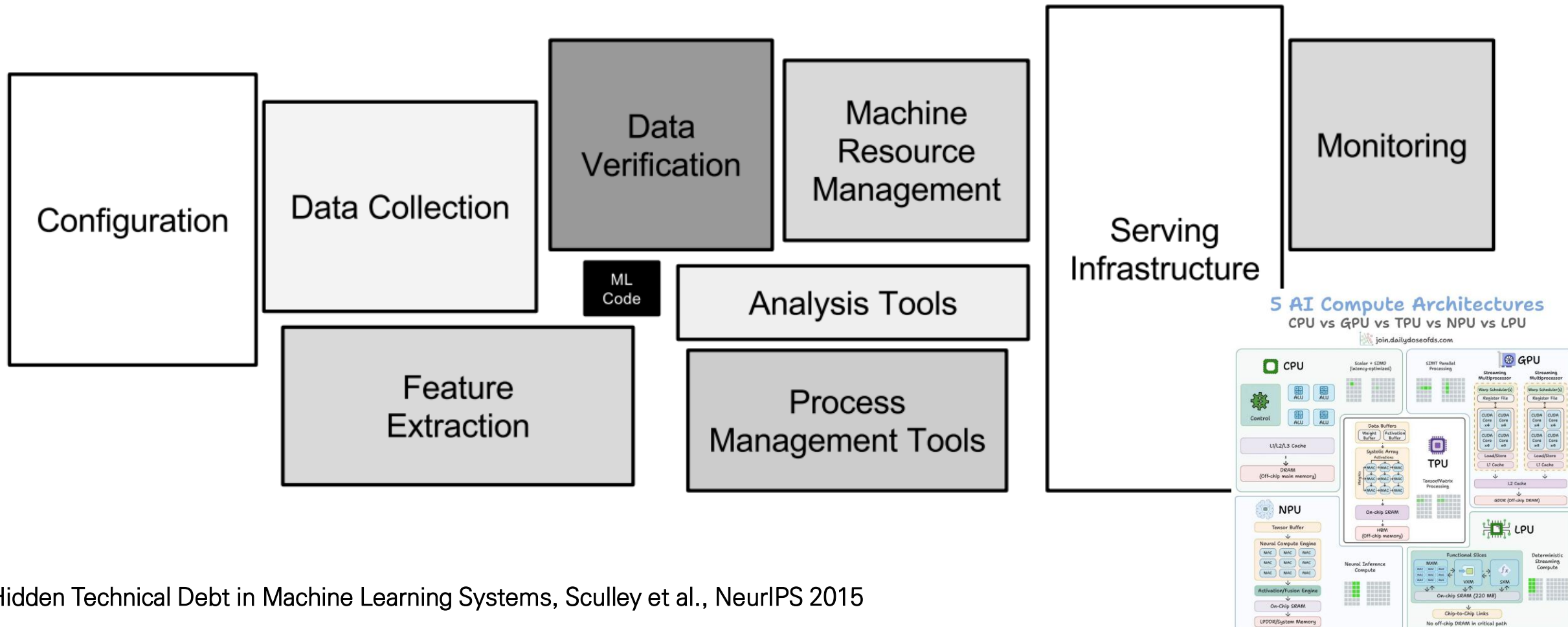


AI Computing – Full Stack AI Expert

A small black square icon with the text "ML Code" in white, centered within the square.

ML
Code

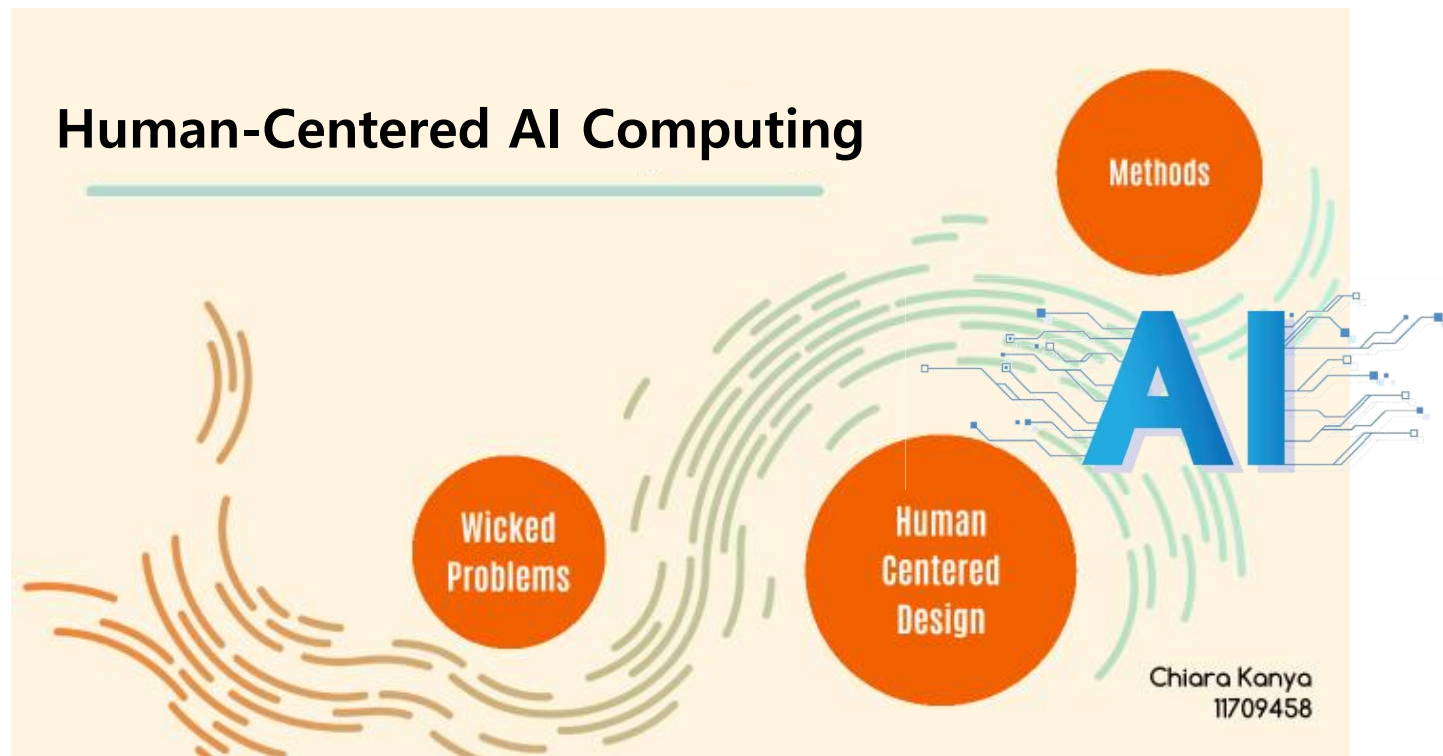
AI Computing – Full Stack AI Expert



KAIST AI Computing Department Vision

Human-Centered AI Computing

An interdisciplinary field focused on creating AI computing systems that support and enhance human activities.



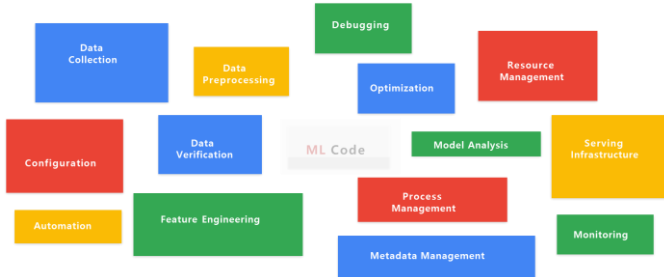
AI IN CS



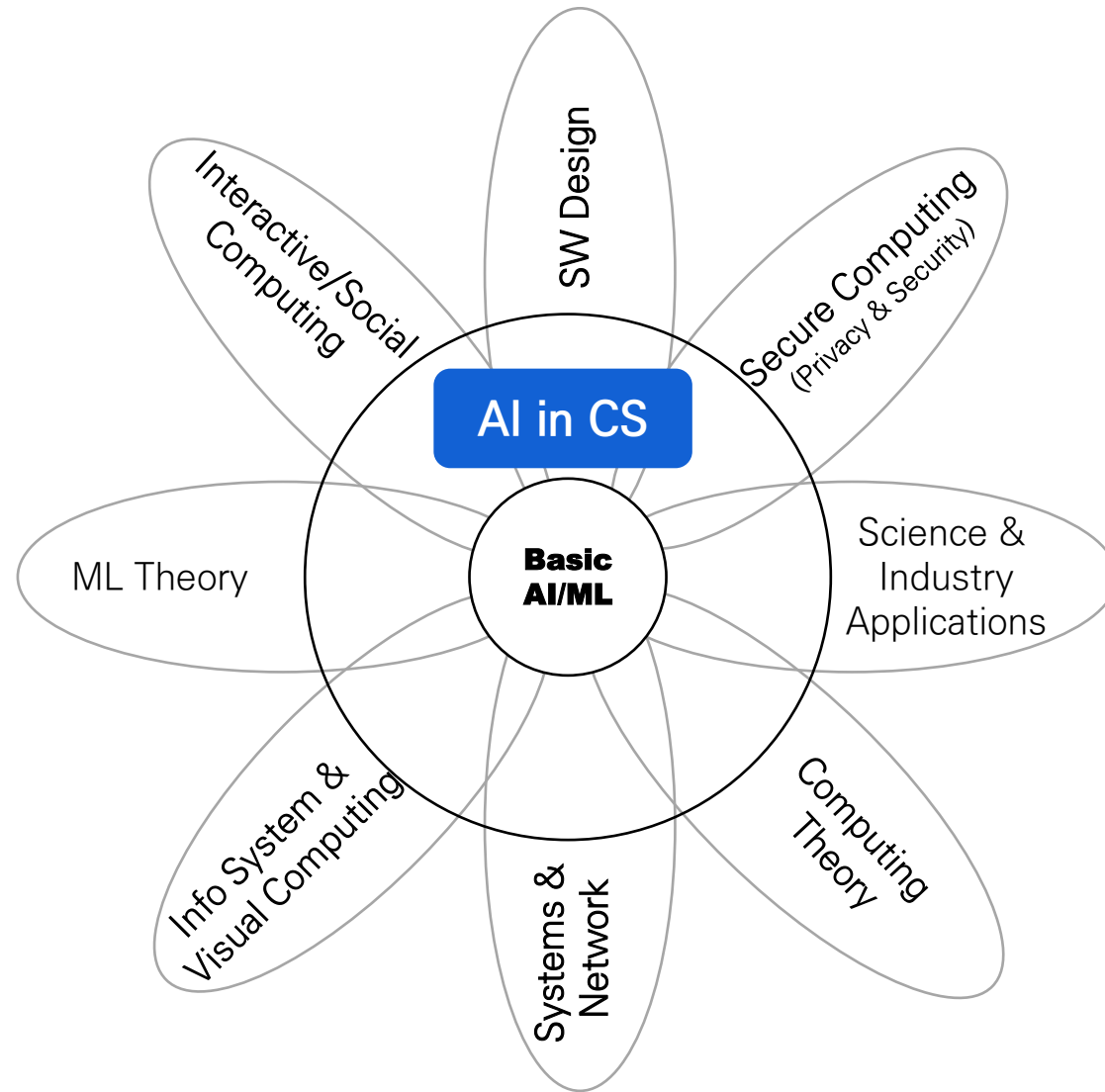
<https://lastfuturist.com/what-is-human-centric-ai/>

Human-Centric AI Computing

* Kim Jaechul
Grad School of AI



Beyond ML Code –
AI Systems & Applications
“Hidden Technical Debt in Machine Learning
Systems, Sculley et al., NeurIPS 2015”



* AX Dept.
* Grad School of
Data Science

* AI Systems Dept.

AI COMPUTING

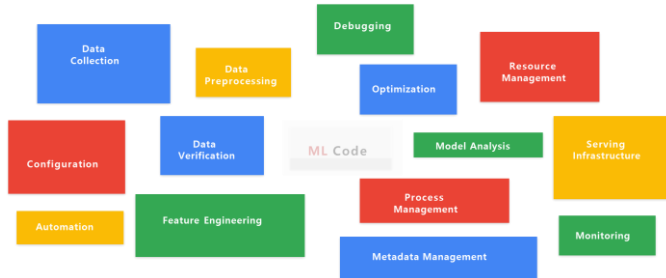
SPECIALIZED PROGRAM FOR
HUMAN-CENTERED AI COMPUTING



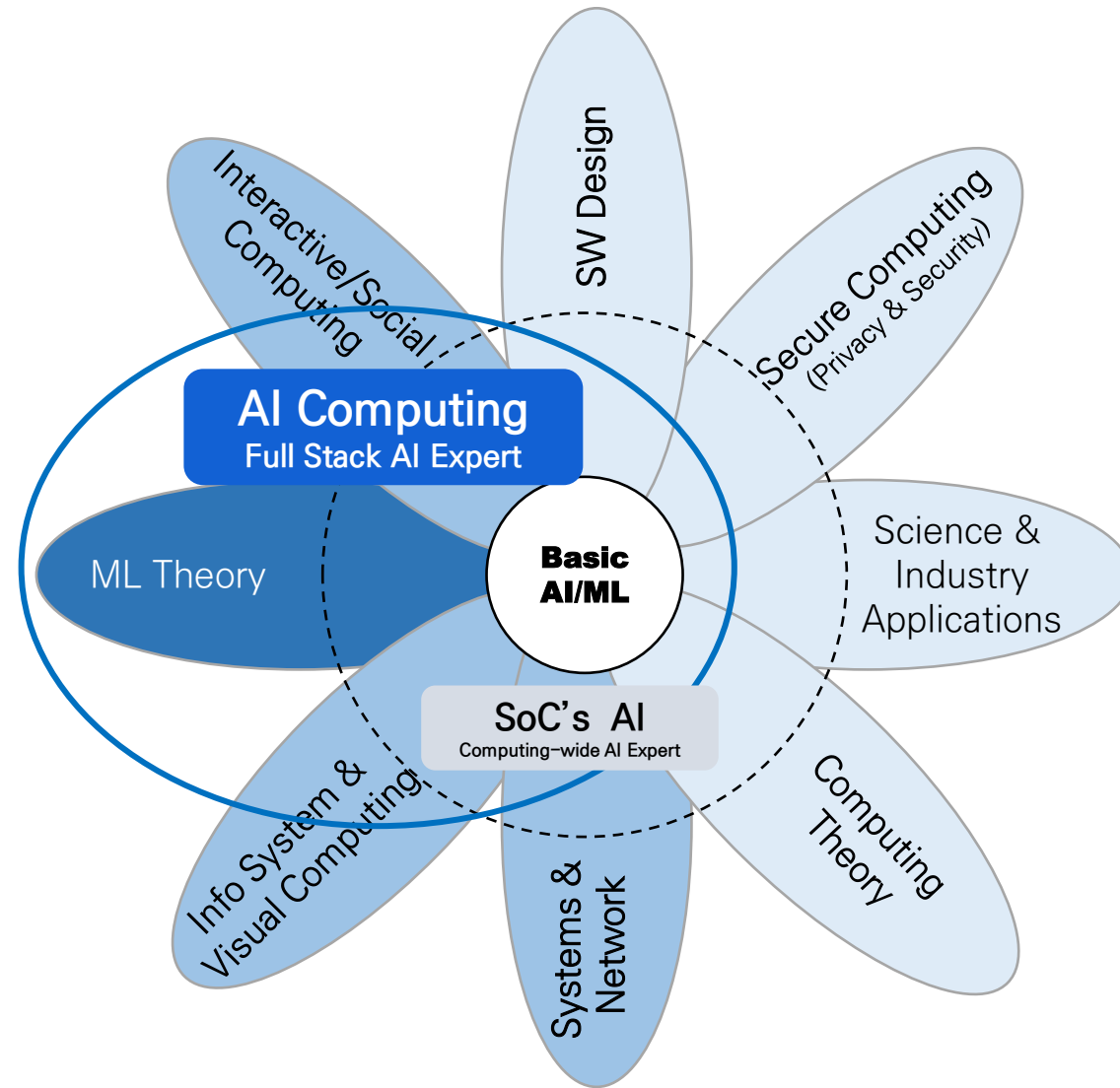
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Human-Centric AI Computing

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“Hidden Technical Debt in Machine Learning
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* AX Dept.
* Grad School of
Data Science

* AI Systems Dept.

Global Leading Faculty of 30 Members



Uichin Lee
CORE FACULTY,
CHAIR



Minhyuk Sung
CORE FACULTY



Sungjin Ahn
CORE FACULTY



Seunghoon
Hong
CORE FACULTY



Joyce Jiyoung
Whang
CORE FACULTY



Noseong Park



Daehyung Park



Jongse Park



Jinah Park



Soeul Son



In-Young Ko



Youngjin Kwon



Daeyoung Kim



Min-Soo Kim



Min H. Kim



Joseph Seering



Junehwa Song



Tae-Hyun Oh



Alice Oh



Shin Yoo



Juho Kim



Tae-Kyun Kim



Hyunwoo J. Kim



Sukyoung Ryu



Sue Bok Moon



Sung-Eui Yoon



Geehyuk Lee



Jae-Gil Lee



Sungho Jo

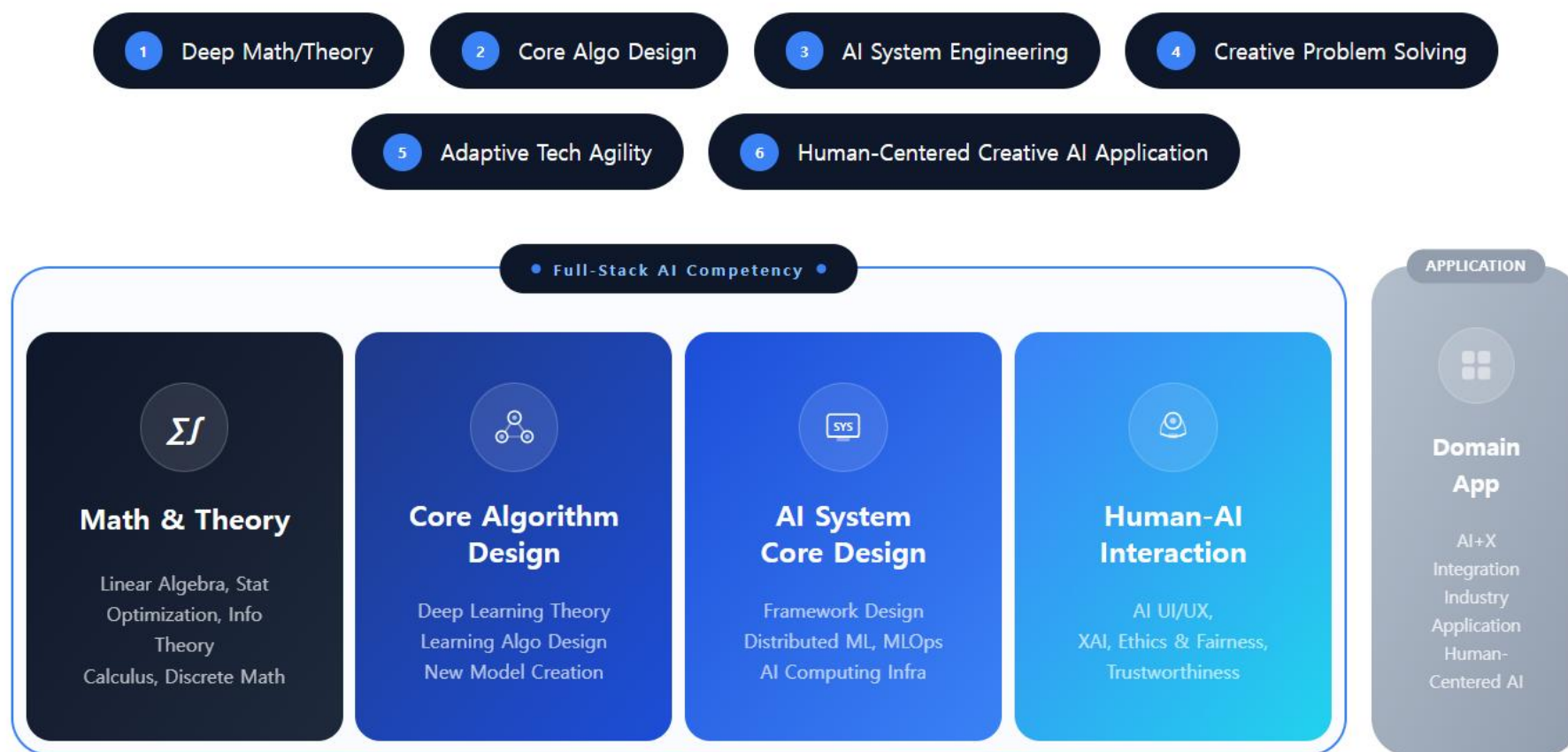


Jun Han

<https://aic.kaist.ac.kr>

AI-Native Education for Future Leaders

Educating AI-Native leaders for the era of Human-Centered AI Computing



AI-NATIVE CORE COMPETENCIES

Responsible AI
AI Design Grounded in Ethics and Philosophical Principles

Domain Application
AI+X Solution Across Industry, Science, and Public Sectors

Next-Generation AI
Leading Innovation in LLM, Generative, and Agentic AI

AI System Engineering
GPU programming, MLOps & distributed AI systems optimization

AI Design Capability
Deep understanding of AI principles and designing new model

Undergraduate Degree

- Major, Double Major, Minor Requirements
 - Major (전공) – 44 credits
 - 12 credits of required major courses + 32 credits of major electives
 - Double Major (복수전공) – 40 credits
 - 12 credits of required major courses + 28 credits of major electives
 - Double-counting 6 credits (CS.30706 Machine Learning, CS.30701 Deep Learning)
 - Minor (부전공) – 21 credits
 - 12 credits of required major courses + 9 credits of major electives
- Curriculum – Major Courses
 - AI Mathematics: Mathematics for AI
 - AI Foundations: Introduction to AI, Machine Learning, Deep Learning
 - AI Algorithms: Reinforcement Learning, Graph Neural Networks, Natural Language Processing, Computer Vision
 - AI Systems: GPU Programming, Systems for AI
 - AI + Human: Human–AI Interaction, Wearable Interfaces
 - AI Philosophy & Ethics: Responsible & Trustworthy AI
 - AI Capstone Project

Department of AI Computing | Undergraduate Curriculum Roadmap



Computing Basics

MAS.10009
Introduction to Linear Algebra

AIC.20002
Discrete Mathematics

MAS.20050
Probability and Statistics

AIC.20003
Data Structure

AIC.30000
Introduction to Algorithms

AIC.30002
AI-Assisted Software Engineering

AIC.30201
Introduction to Data Science

AIC.30001
Optimization Methods for AI



AI Foundation

AIC.10001
Elements of AI

AIC.20000
Introduction to Artificial Intelligence

AIC.20004
AI Philosophy: Ethics and Law

AIC.20101
Introduction to Deep Learning

AIC.20100
Machine Learning

AIC.30101
Introduction to Bayesian Machine Learning

AIC.20001
Mathematics for AI

AIC.30100
Introduction to Statistical Learning Theory



AI Algorithm

AIC.30300
Introduction to Large Language Models

AIC.40100
Graph Machine Learning and Mining

AIC.30102
Introduction to Reinforcement Learning

AIC.40300
Machine Learning for Natural Language Processing

AIC.30400
Introduction to Computer Vision

AIC.40400
Machine Learning for Computer Vision

AIC.30103
Introduction to Generative Models

AIC.40401
Machine Learning for 3D Data

AIC.40101
Brain-Inspired Machine Intelligence

Course Completion Requirements

- Major** | Required: 12+ & Elective: 32+
- Minor** | Required: 12+ & Elective: 9+
* No credits from the same course will be doubly counted to satisfy major and minor department requirements.
- Double Major** | Required: 12+ & Elective: 28+
* Up to 6 credits can be doubly counted to satisfy both major department requirements.



College of AI
AI대학



AI + Human

AIC.40501
Multi-Agent Engineering for Social Computing

AIC.40503
Responsible AI



AI System

AIC.20500
Introduction to Human-Computer Interaction

AIC.30500
Human-AI Interaction

AIC.40502
Wearable User Interface

AIC.20600
System Programming

AIC.30602
Operating Systems and Lab

AIC.40600
System for Artificial Intelligence

AIC.30601
GPU System Programming

AIC.30603
Computer Organization



Capstone Project

AIC.30901
Capstone Project I

AIC.40901
Capstone Project II

- * This course tree is tentative and please see the up-to-date info at the home page
- * Until Fall 2026, please take substitute courses in CS
- * As of Fall 2026, students should take AIC code courses
- * See the details: aic.kaist.ac.kr

Graduate Program in AI Computing Department

- AI Computing: Admitting approximately 20 MS + 5 PhD students (Fall 2026)
- Proposed Graduate Curriculum
 - Advanced AI Theory & Algorithms: AI & ML, Reinforcement Learning, Advanced ML
 - Advanced AI Systems: Distributed AI, Large-Scale Model Training, System Architecture Design
 - Resource-Efficient AI Technologies: Edge AI, Model Compression
 - AI + Human (Advanced): Advanced Human-AI Interaction
 - AI Ethics & Safety: AI Model Safety Evaluation, Privacy and Security Issues
 - Emerging AI Topics: Large Language Models, Vision-Language Models, Generative Models, Multimodal Learning, Retrieval-Augmented Generation (RAG)
 - Emerging Research Areas: Agentic AI, Physical AI
 - AI Research Seminars: Latest research presented at top-tier conferences in AI, ML, NLP, and Computer Vision
 - Industry Internship (4-6 months): Hands-on experience applying research outcomes to real-world problems in leading AI companies

Q&A

- CS ⇔ AIC (Major/Double-Major) – Double Counting
 - Two courses are eligible (전산 → AIC): CS.30706 Machine Learning, CS.30701 Deep Learning
 - Courses from other departments will be considered in the future
- Differences between AI Specialized Minor ([AI 특별지정전공](#)) vs. AI Computing Minor
 - AI Specialized Minor: 18 credits = 6 required + 12 electives
 - Required Major (전공필수): Programming Basics + AI Basics
 - Designed Electives (지정선택): NLP, CV, Robotics, Advanced ML, Data Science, AI in Society
 - Electives (선택과목): X+AI, AI Basics
 - AI Computing Minor: 21 credits = 12 required + 9 electives
 - Required Major (전공필수): AI Philosophy and Ethics, Intro to AI, ML, Deep Learning
 - Major Electives (전공선택): Any major courses in AI Computing
 - More specialized for Core AI Computing subjects
- Grad School Minor (AI대학 대학원 부전공 프로그램): No plans yet



NO MORE TIME TO EXPLAIN, HOP IN!
더 설명할 시간 없어, 어서 타!



AI-Native Education for Future Leaders

Educating AI-Native leaders for the era of Human-Centered AI Computing

<https://aic.kaist.ac.kr>